

Modeling Natural Patterns: An Exploration of Reaction-Diffusion Systems

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ABSTRACT

Patterns in nature from the stripes of fish to the arrangement of vegetation in arid landscapes often emerge from underlying physical and chemical processes. In this paper we explore how reaction-diffusion systems, particularly the Turing instability framework, gave rise to such patterns through mechanisms of local activation and long-range inhibition. We begin by mathematically deriving the conditions under which a homogeneous steady state becomes unstable, leading to spontaneous pattern formation. Focusing on the Gierer–Meinhardt model, we present simulations that demonstrate the development of spots and stripes, along with how the introduction of a saturation term influences the resulting dynamics. Using tools like VisualPDE, we visualize how parameter variations can shift patterns and connect these theoretical models to real-world examples such as fish skin pigmentation and ecological vegetation bands. Our findings illustrate the power and flexibility of reaction-diffusion systems in modeling complex, self-organizing structures found in both biology and the environment.

1. INTRODUCTION

From fish stripes and leopard spots to sand dunes and vegetation, patterns emerge out of nowhere and prevail in nature. Understanding how such ordered structures arise in initially uniform, steady-state environments has fascinated scientists for decades. One such scientist, Alan Turing, pioneered the science behind chemical reaction and diffusion, showing how chemicals can destabilize a homogeneous state and give rise to spatial patterning (A. M. Turing 1952). These reaction-diffusion systems are the foundation of a breadth of pattern-forming phenomena—in chemistry, biology, ecology, and beyond.

In a reaction-diffusion system, chemical agents spread throughout space and interact with each other. After certain thresholds are met—when the agents’ diffusion rates differ enough and the reaction kinetics reach a level of stability—chemical diffusion transforms from a homogeneous mechanism to a pattern-generating one. This mechanism, now known as *Turing instability*, has been used to explain stripe and spot formation on animal skins, vegetation banding in arid landscapes, and even nano-scale patterning in chemical gardens (J. Murray 2003).

In this paper, we review the theory of reaction-diffusion pattern formation and illustrate key mathematical principles behind Turing instability. Then, we apply these ideas to the Gierer–Meinhardt model, reproducing patterns and introducing saturation that enriches our models. This report aims to provide a clear mathematical understanding of linear stability analysis while highlighting the rich tapestry of patterns made possible through reaction-diffusion dynamics.

2. BACKGROUND

The reaction-diffusion equation is a mathematical way to describe how chemicals move and react in space and time. It’s used to model how patterns (such as animal stripes or spots) form as a result of chemical interactions. From this result we can mathematically prove Turing’s instability, which shows that diffusion can destabilize a uniform steady state under certain conditions. This is what leads us to patterns seen in nature like stripes and spots on animals. To prove this, we will consider a 2-variable reaction-diffusion system.

2.1. Reaction-Diffusion Equation

To begin, we can look at the vector form of the reaction diffusion equation:

$$\frac{\partial \mathbf{c}}{\partial t} = \mathbf{f}(\mathbf{c}) + D \nabla^2 \mathbf{c} \quad (1)$$

Where:

$$\mathbf{c}(\mathbf{r}, t) = \begin{bmatrix} A(\mathbf{r}, t) \\ B(\mathbf{r}, t) \end{bmatrix} \quad (2)$$

is the vector of concentrations, $A(\mathbf{r}, t)$, $B(\mathbf{r}, t)$ is concentration of given species A, B at location $\mathbf{r}$ and time t .

$$\mathbf{f}(\mathbf{c}) = \begin{bmatrix} f(A, B) \\ g(A, B) \end{bmatrix} \quad (3)$$

are the nonlinear reaction terms, which describe how A and B change due to chemical reactions between them.

$$D = \begin{bmatrix} D_A & 0 \\ 0 & D_B \end{bmatrix} \quad (4)$$

is the diagonal matrix of diffusion coefficients, representing the rate at which each species diffuses.

∇^2 is the Laplacian operator.

Substituting into our given diffusion equation we end up with our component-wise system:

$$\frac{\partial A}{\partial t} = f(A, B) + D_A \nabla^2 A \quad (5)$$

$$\frac{\partial B}{\partial t} = g(A, B) + D_B \nabla^2 B \quad (6)$$

2.2. Homogeneous Steady State

In order to find a homogeneous steady state, we want to find a constant solution:

$$\mathbf{c}_0 = \begin{bmatrix} A_0 \\ B_0 \end{bmatrix}, \quad \nabla^2 \mathbf{c} = 0 \quad (7)$$

This means the system does not change in time and the concentrations are uniform in space. Plugging into the original equation we get:

$$0 = \mathbf{f}(\mathbf{c}_0) \quad (8)$$

Which means:

$$f(A_0, B_0) = 0, \quad g(A_0, B_0) = 0 \quad (9)$$

This is our standard fixed point; if there are no perturbations it will stay constant.

2.3. Small Perturbations

Suppose we disturb our system with a small perturbation, δ , such that:

$$\mathbf{c}(\mathbf{r}, t) = \mathbf{c}_0 + \delta \mathbf{c}(\mathbf{r}, t) = \begin{bmatrix} A_0 \\ B_0 \end{bmatrix} + \begin{bmatrix} \delta A(\mathbf{r}, t) \\ \delta B(\mathbf{r}, t) \end{bmatrix} \quad (10)$$

where δA and δB are the small deviations.

Substituting into the original PDE gives:

$$\frac{\partial(\delta \mathbf{c})}{\partial t} = \mathbf{f}(\mathbf{c}_0 + \delta \mathbf{c}) + D \nabla^2(\delta \mathbf{c}) \quad (11)$$

2.4. Linearization

We linearize $\mathbf{f}(\mathbf{c})$ around the steady state using a first-order Taylor expansion:

$$\mathbf{f}(\mathbf{c}_0 + \delta\mathbf{c}) = \mathbf{f}(\mathbf{c}_0) + J\delta\mathbf{c} \quad (12)$$

Since $\mathbf{f}(\mathbf{c}_0) = 0$, we get:

$$\frac{\partial(\delta\mathbf{c})}{\partial t} = J\delta\mathbf{c} + D\nabla^2\delta\mathbf{c} \quad (13)$$

Where J is the Jacobian matrix evaluated at $\mathbf{c}_0 = (A_0, B_0)$:

$$J = \begin{bmatrix} \frac{\partial f}{\partial A} & \frac{\partial f}{\partial B} \\ \frac{\partial g}{\partial A} & \frac{\partial g}{\partial B} \end{bmatrix} = \begin{bmatrix} f_A & f_B \\ g_A & g_B \end{bmatrix} \quad (14)$$

2.5. Fourier Analysis

We now analyze our linearized PDE system using Fourier analysis. Consider a solution of the form:

$$\delta\mathbf{c}(x, t) = \mathbf{u}e^{\lambda t} \cos(kx) \quad (15)$$

Then:

$$\nabla^2\delta\mathbf{c}(x, t) = -k^2\delta\mathbf{c}(x, t) \quad (16)$$

Substituting into our PDE gives:

$$\lambda\mathbf{u} \cos(kx) = J\mathbf{u} \cos(kx) - Dk^2\mathbf{u} \cos(kx) \quad (17)$$

Canceling $\cos(kx)$ from both sides, we get:

$$\lambda\mathbf{u} = (J - k^2D)\mathbf{u} \quad (18)$$

This is an eigenvalue problem that determines how much a system grows or decays. If any eigenvalue $\lambda(k)$ has a positive real part, the homogeneous steady state is unstable and pattern formation may occur. Lastly, we can examine under what conditions $\lambda(k)$, the eigenvalues of the matrix $J - k^2D$, become positive for some spatial frequency $k > 0$. This means the corresponding Fourier mode will grow over time and destabilize the system. This is where Turing instability occurs. If the steady state is stable without diffusion, such as when $k = 0$, and λ is an eigenvalue of J , the system requires both real parts of the eigenvalues of J to be negative. This stability condition holds when:

$$f_A + g_B < 0, \quad f_A g_B - f_B g_A > 0 \quad (19)$$

When we introduce diffusion, the matrix becomes $J - k^2D$, and $\lambda(k) > 0$ for some $k > 0$. This instability can happen when the diffusion rates are significantly different, and normally happens when the inhibitor diffuses much faster than the activator. This is when D_B is far greater than D_A . This shows that even though the system is stable without diffusion, and adding diffusion can cause certain spatial modes to grow. This growth is what leads to the formation of spatial patterns, which is key in Turing's work and in the instability proof.

2.6. Activator-Inhibitor Model

This section dives into the activator-inhibitor model: a type of reaction-diffusion system in which two chemical species interact with each other, driving pattern formation. For the sake of simplicity, the chemical species will be defined as species A and B. In this model, species A acts as the activator—it determines how much of itself and the inhibitor, species B, is produced. Mathematically, this behavior is captured by the partial derivative of the activator's reaction rate with respect to itself:

$$\frac{\partial F}{\partial A} > 0 \quad (20)$$

In contrast, the inhibitor suppresses the production of the activator, typically modeled by:

$$\frac{\partial F}{\partial B} \leq 0 \quad (21)$$

The relationship between species A and B forces a specific interaction to occur. While the activator reinforces itself in small regions, the inhibitor diffuses faster and faster and prevents the activator from spreading elsewhere. Eventually, the inhibitor diffuses faster than the activator, causing a great imbalance which destabilizes the otherwise homogeneous environment. This unique imbalance is the foundation of Turing patterns since it turns its environment into organized structures like spots or stripes.

This relationship was first expanded upon in Alan Turing’s theory of morphogenesis (?) where he proposed that certain biological patterns arise from chemical interactions in reaction-diffusion equations. The UCSD lecture notes on nonlinear dynamics emphasize that Turing’s central insight was that diffusion, usually seen as a stabilizing force, can instead create instability when paired with specific reaction kinetics (D. Arovav 2023). Turing’s criteria for instability involve the eigenvalues of the linearized system becoming positive for certain spatial frequencies—meaning certain perturbations will grow rather than decay over time.

J.D. Murray further developed and formalized these ideas in Mathematical Biology Volume II, where the activator-inhibitor model is used to explain a variety of biological phenomena. These include animal coat patterns (e.g., zebra stripes, leopard spots), digit formation, and even feather or hair follicle arrangement. Murray outlines the mathematical conditions for pattern formation and emphasizes how changing diffusion coefficients and nonlinear reaction terms can lead to drastically different spatial outcomes.

In summary, the activator-inhibitor model captures the essence of Turing’s mechanism: simple local rules, when coupled with differential diffusion, can give rise to rich and complex patterns observed throughout nature.

3. GIERER-MEINHARDT MODEL VISUALIZATION

Over the past few decades, a variety of two-component reaction-diffusion models have been proposed and studied in detail. Among them, the Gierer–Meinhardt model (A. Gierer & H. Meinhardt 1972) captures the essential features of *local activation* and *long-range inhibition*, and serves as a paradigmatic system for numerical simulation and theoretical analysis. Modern computational tools, such as VisualPDE (B. J. Walker et al. 2023a), allow for real-time visualization of the intricate dynamics that emerge from simple mathematical rules, making it possible to explore parameter regimes that yield spots, stripes, rings, and other complex motifs.

This is a visualization of the Gierer–Meinhardt Model from VisualPDE (B. J. Walker et al. 2023a). Taken at various points in time, this whole process took about 15 seconds to turn one perturbation as seen in Figure 1 into a spot-like pattern across the screen as seen in Figure 2. Eq 22 & 23 represents our model where we take $a, b, c > 0$ and $D > 1$ (B. J. Walker et al. 2023b). Eq 22 is the activator in this model while Eq 23 represents the inhibitor.

$$\frac{\partial u}{\partial t} = \nabla^2 u + a + \frac{u^2}{v} - bu \quad (22)$$

$$\frac{\partial v}{\partial t} = 100\nabla^2 v + u^2 - cv \quad (23)$$

This is a visual representation of how a Turing Instability can emerge which in result creates patterns spontaneously with respect to the differing diffusion rates. The key characteristics to this model is the local activation and long-range inhibition.

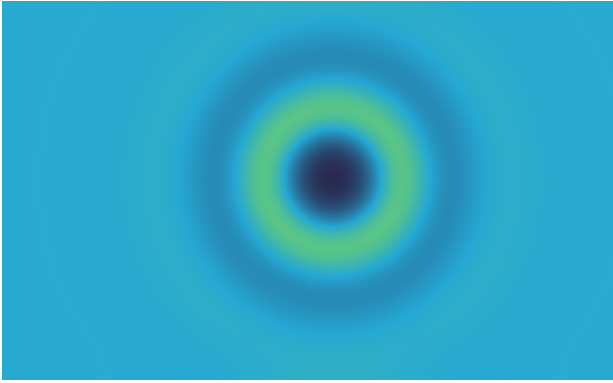


Figure 1. 2 seconds after a perturbation was introduced into the system. See how the perturbation created a dip and the wave the dip created will turn into a peak.

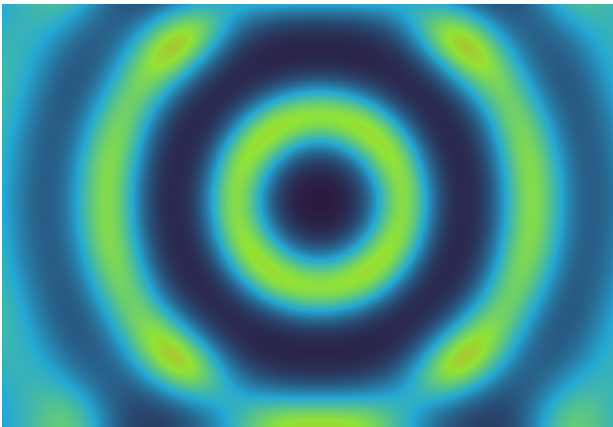


Figure 3. 5 seconds after a perturbation was introduced into the system. There are now rings forming around the initial perturbation. This is how the model spreads across the screen.

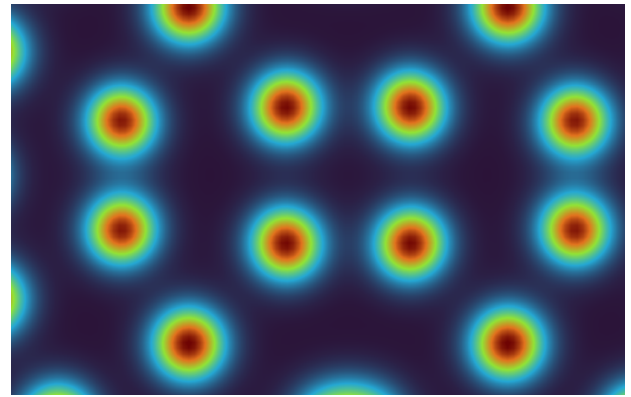


Figure 2. 14 seconds after the initial perturbation was introduced into the system. This is what the system looks like once it has stabilized. You can see here how the spots are formed following this model.

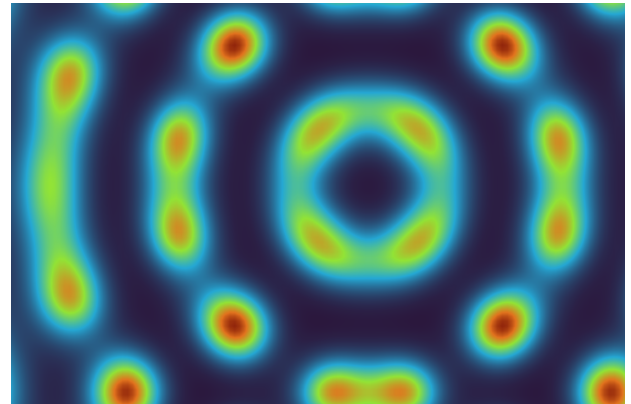


Figure 4. 8 seconds after the initial perturbation was introduced into the system. You can now see a dot forming in each quadrant of the initial circle, and also in the surrounding rings. These are the spots forming into these singular points.

140 The local activation involves the activator to increase where it is already high. This is noticeable in Figures 1 and
 141 3 where the peaks continue to increase and not change direction as time moves forward. The perturbation created a
 142 dip in the model and that dip created a circular peak around it. This increased more and more until it created four
 143 spots in each quadrant of the circle, this can be seen in Figure 5.

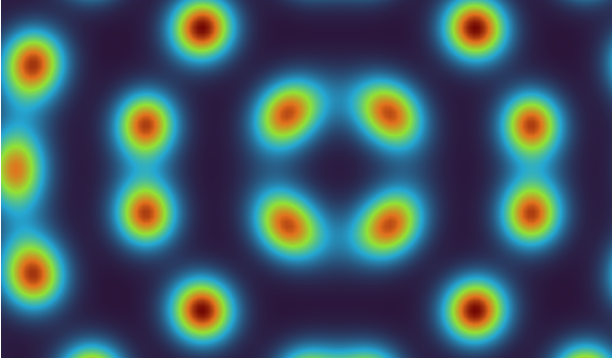


Figure 5. 10 seconds after the initial perturbation. Now the spots are more prevalent

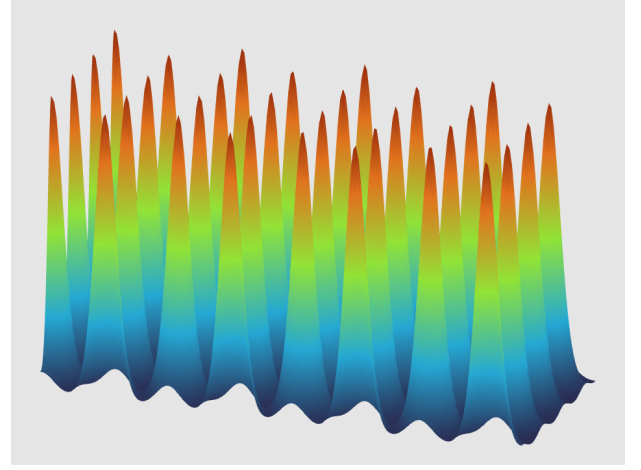


Figure 6. Here is a plane view of the model after the spots have formed. Notice how the spots are peaks in the model and the space in between them are deep dips.

Long-range activation causes the inhibitor to spread faster while also suppressing activation in the regions nearby. Notice in Figure 2 the blue area around the spots are actually dips, this would be the result of the long range activation suppressing the regions around the spots.

These characteristics together give way to spatially periodic patterns. Figure 6 shows the spots from a 3D view. The spots are peaks while the regions in between are dips in the model. We can change the pattern from spots to stripes by making a slight adjustment to the activator function (Eq 24)

3.1. Introducing a Saturation Constant

Now we are still working with the Gierer-Meinhardt model, but we have made a slight change to the model. In Eq 24 we made a change to the self-activation term $\frac{u^2}{v}$ to include $\frac{1}{(1+Ku^2)}$ to the term. The inhibitor function Eq 25 is left untouched as well as making sure $a, b, c > 0$ and $D > 1$. While making sure that $K > 0$ this will be our saturation constant (B. J. Walker et al. 2023b).

$$\frac{\partial u}{\partial t} = \nabla^2 u + a + \frac{u^2}{v(1+Ku^2)} - bu \quad (24)$$

$$\frac{\partial v}{\partial t} = D\nabla^2 v + u^2 - cv \quad (25)$$

In Figure 7 instead of a dot we made a diagonal line across the screen to show how stripes can emerge from this model. Figure 8 shows what the model looks like once the pattern has fully spread across the screen. This model took 10 seconds to fully form the stripes across the screen. Comparing it to the previous model, where it favored a spot like pattern, this model is able to produce stripes thanks to the K constant. One thing to note about the K constant for this model large values for K will not result in creating a Turing pattern. For small values it will behave like the previous model and create a spot-like pattern, since it would return to the $\frac{u^2}{v}$ form. For this model we set $K = 0.003$; setting it to $K = 0.001$ would cause the stripes to begin forming spots.

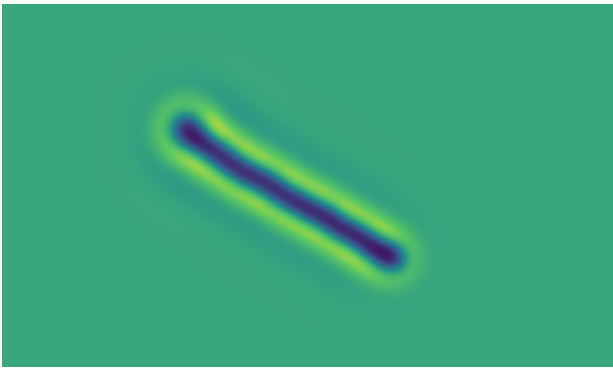


Figure 7. The initial perturbation to the system. Done in a straight line to demonstrate this model's ability to form stripes

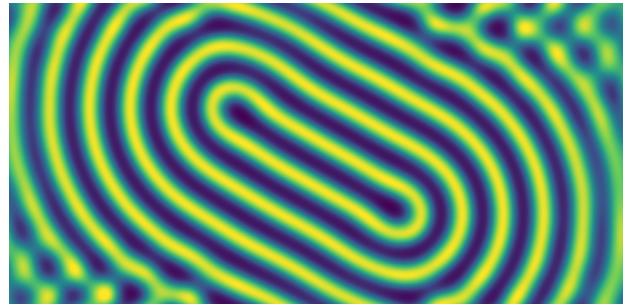


Figure 8. 10 seconds after the initial perturbation. You can how the stripe pattern radiates across the screen.

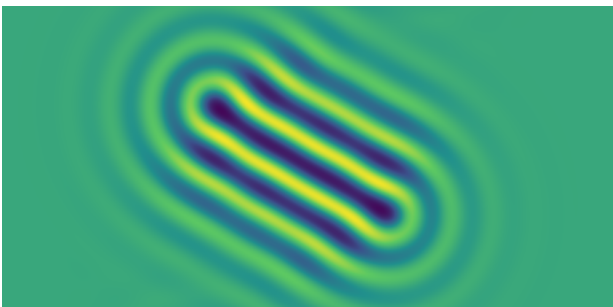


Figure 9. 3 seconds after the initial perturbation to the system. You can now see the pattern radiating outward from the initial zone.

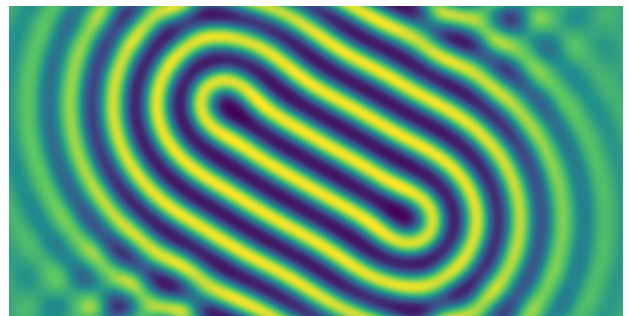


Figure 10. 7.5 seconds after the initial perturbation to the system. See how the outer rims of the model begin to saturate into more apparent stripes now.

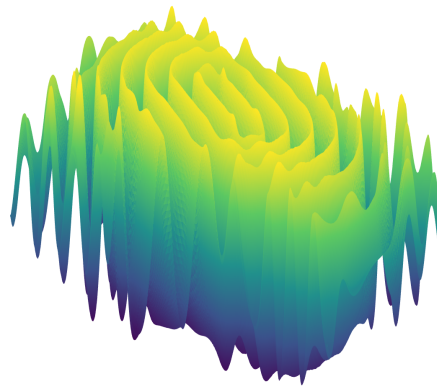


Figure 11. 3D view of the model after 10 seconds.

Figure 11 is a view of the model in a 3D plane. Notice how the peaks and dips are similar to that of the previous model, where the blue areas are actually dips and the lighter colors are peaks within the model. Now instead of spots, these are stripes forming. From a 3D view this is similar to looking at a canyon of various peaks and dips throughout.

4. REAL-WORLD APPLICATIONS

Stepping back from the equations and theory, it's striking how reaction-diffusion ideas show up all around us. Below are a few areas where the same math and models we've discussed shed real light on natural and engineered patterns.

171 Diffusion patterns can be seen throughout nature as evident in Figure 12. See how the stripes form in a pattern
 172 similar to the models seen in the previous section.



Figure 12. Stripes on a tuna seen from a fish market. Excuse the flies around it.

173 The Gierer–Meinhardt model explains how fish such as the Regal Angelfish, as seen in Figure 13, develop their
 174 horizontal stripe patterns through a reaction–diffusion mechanism involving two interacting substances: an activator
 175 and an inhibitor. The activator promotes its own production and the production of the inhibitor, while the inhibitor
 176 spreads more rapidly and suppresses the activator in the surrounding areas. On the growing skin of the fish, this local
 177 self-enhancement and long-range inhibition leads to the formation of regularly spaced stripes. As a result, the model
 178 predicts how pigment cells differentiate and organize into stable, evenly spaced horizontal bands, matching the visually
 179 striking patterns observed in many reef fish.



Figure 13. Stripes of a Regal Angelfish.

180 Out in arid drylands, bundles of plants known as “tiger bush” form neat stripes or patches along slopes and plains.
 181 In these conditions, the plants and water play the roles of activator and inhibitor: plants use up resources locally
 182 (activation) but deplete moisture farther out (inhibition). Field data actually match the pattern wavelengths our
 183 stability analysis predicts, and adding a saturation term helps explain why some regions shift from isolated shrubs to
 184 continuous green belts when the climate changes (M. Rietkerk et al. 2004). This can be seen in Figure 14.

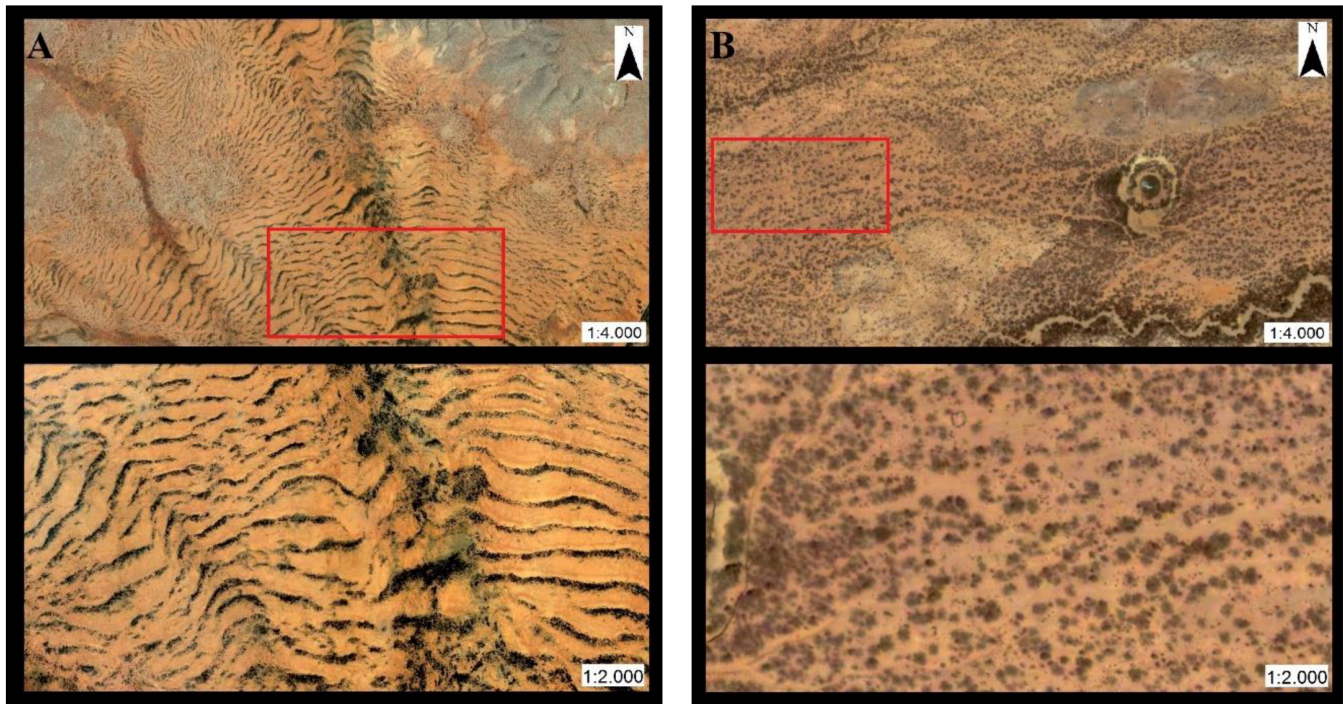


Figure 14. "Tiger Bush" plant patterning in arid ecosystem.

5. CONCLUSION

In summary, this report explored how reaction–diffusion systems can turn homogeneous systems into the many natural patterns we see today. All it takes is to tweak a few parameters—like diffusion rates or reaction terms—to achieve a breadth of different outcomes, from stable steady states to complex patterns like spots and stripes.

Using the Gierer–Meinhardt model, we were able to actually see how small perturbations evolve over time into organized structures. The idea of local activation and long-range inhibition played a big role here, and it became clear how important the balance between those two effects is. Adding in a saturation term gave us even more flexibility, letting us shift the system from spots to stripes just by changing one part of the equation.

In the end, reaction–diffusion models give us a useful way to understand how patterns in nature might come about—not just in theory, but in actual biological and chemical systems. Even though the math can seem a bit abstract at times, the results are surprisingly intuitive once you see them in action.

APPENDIX

6. CHATGPT USAGE

This is a following reponses to our individual uses of ChatGPT throughout this project:

6.1. *Raynell's Response*

ChatGPT was used throughout this paper, mainly for its use of LaTeX formatting. This was very helpful as we had group members inexperienced with LaTeX and using ChatGPT allowed them to format their sections with little to no changes from the people who formatted this report fully. Another way we used Chat is helping us structure and form our paper. This helped straighten out our ideas and made it clear on what we needed to do for the paper.

6.2. *Michael's Response*

For the Turing Instability proof portion of the paper, Chat GPT was used to help me structure the structure of the reaction diffusion equation and how it models pattern formation. It also helped me structure the formatting of the entirety of the proof, by changing it into AASTeX style that we had decided was the best format for this research paper. It was also used to translate each equation I had written out into formatting for Latex for presentability. It

also broke down some of the ideas in the proof that I was having trouble understanding, and provided some deeper insight into each part of the proof. It was lastly used to confirm the overall accuracy of the proof, making sure my ideas had lined up correctly and were presented in a format that makes sense.

6.3. *Isabella's Response*

To support this report I used ChatGPT to collaborate. I started by writing a rough outline of my ideas and trying to expand on them as much as possible. I uploaded the outline, the context in the overall report, and the two sources I wanted to pull from. Then I asked for a more detailed and technical assessment of the information. The AI was able to not only expand on what was already there, but it also made it more accurate and easier to understand. Another helpful feature was that the tone of the writing became more academic so it works better for a research paper.

6.4. *Xander's Response*

Artificial intelligence has become a useful tool for saving time and expediting work. For this assignment, I occasionally felt "in the dark" on where to begin my research and how to determine which sources made valuable references. With the help of ChatGPT, I could simply explain a topic and ask for credible sources relative to what I was researching. Using AI to help me find sources saved me from a multitude of busy work, and allowed me to spend more time working on meaningful tasks—reviewing sources, drafting the report, and the like. In other words, ChatGPT became the librarian of the internet; Just make sure you verify its sources before using them.

Additionally, AI is a smart writer that I've used to boost my formatting. The tech has exceptional knowledge on diction, line of reasoning, the weight of words—all of which are key traits in making a clear, cohesive report. Because of this, I would sometimes ask ChatGPT to give feedback on my writing and share areas that it thought I could improve. In short, ChatGPT helped me think critically about my writing and taught me how to format my work in an impactful way.

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